Yunus Emre Danabaş
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EDUCATION

Sabancı University, Istanbul, Turkey — 2020 – 2026 (Expected)
B.Sc. in Mechatronics Engineering — *Graduated June 2025* — CGPA: 3.77/4.00
B.Sc. in Electronics Engineering (*Double Major*) — *Expected Graduation: June 2026* — CGPA: 3.77/4.00

- Combined specialization in robotics, control, and embedded systems across both majors
- Selected coursework: Kinematics and Dynamics of Machines; Introduction to Robotics; Autonomous Mobile Robotics; Deep Learning for Robot Control

RESEARCH & ENGINEERING EXPERIENCE

As Robotics (COMAU Turkey) — Istanbul, Turkey
Part-time Robotics Engineer — Oct 2025–present

- Designing a custom touch-based grip sensor to enable zero-gravity manual guidance; implementation ongoing
- Developing a handheld motion-teaching device that records end-effector poses/orientations for autonomous replay; implementation ongoing
- Contributing to mechanical integration and control logic across both projects (C++/Python, industrial robot APIs); completed PDL2 training

PRISMA Lab, University of Naples Federico II — Napoli, Italy
Research Intern, Supervised by Prof. Vincenzo Lippiello — Jun 2025–present

- Designing a per-side actuated suspension for a compact lunar rover (roll/heave; passive pitch)
- Building CAD + ROS 2/Gazebo models with URDF/Xacro; posture presets, a hybrid suspension plugin for passive pitch-averaging, and roll-leveling tests on 5–10° ramps; simulation refinement ongoing
- Integrating self-locking linear actuators per side and synthesizing design guidelines from lunar rover, mobile robotics, and suspension literature for low-power, lightweight implementation

MIRMI, Technical University of Munich (TUM) — Munich, Germany
Research Intern, Supervised by M.Sc. Mehmet Can Yıldırım — Jun 2024–Aug 2024

- Investigated crossed-flexure hinge architectures for 6-DoF force–torque sensing to raise moment capacity without sacrificing force resolution
- Built a parametric MATLAB pipeline from Wittrick's crossed-strip theory to explore geometries, loads, and deformation
- Fabricated and instrumented prototypes (SolidWorks, strain gauges) and validated with a computer-vision pipeline (edge detection, RANSAC) against models
- Created Gazebo digital twins for flexure-stage integration to de-risk controller design and pre-hardware tuning

TÜBİTAK BİLGEM — Gebze, Turkey
Robotics Engineering Intern — Sep 2023–Oct 2023

- Completed formal ROS training focused on navigation, SLAM, and multi-robot systems
- Developed and evaluated single- vs multi-robot frontier exploration in Gazebo (TurtleBot3, ROS 1, RViz)
- Implemented real-time map merging and visualization to validate multi-robot coverage

Pubinno Inc. — Istanbul, Turkey
Mechanical Engineering Intern — Jun 2023–Sep 2023

- Designed and 3D-printed rapid prototypes in Onshape to accelerate iteration and fit manufacturing constraints
- Built an automated maintenance prototype for first-line servicing; supported QA/QC with suppliers and documentation

TÜBİTAK BİLGEM — Gebze, Turkey
Intern – Embedded Systems and Digital Design — Jan 2023–Feb 2023

- Developed Python SDR demos with Adalm-Pluto for wireless communication and signal processing

TEACHING EXPERIENCE

Sabancı University — Istanbul, Turkey
Teaching Assistant for Prof. Volkan Patoğlu — Sep 2024–present
Courses: ME312 Analysis and Synthesis of Mechanisms; ME403 Introduction to Robotics

- Led labs/recitations, held weekly office hours, and supervised course projects for 20+ students per semester

PROJECTS

Hand-Steer Sim — Gesture Teleoperation for Mobile Robots Mar 2025–May 2025

- Built a webcam-only gesture teleop system; MediaPipe landmarks drive an MLP (static) and LSTM (steering) in a ROS Noetic/Docker stack
- Reached 99% macro-F1 with 13 ms end-to-end latency at 30 FPS on GPU (25 ms CPU)
- One-command launch publishes fused `/cmd_vel` and starts camera, inference, and Gazebo; released dataset, notebooks, and CPU/GPU Docker images

Passive Walker RL Pipeline (Graduation Project) Oct 2024–May 2025

Supervisors: Prof. Volkan Patoğlu, Dr. Aykut Cihan Satıcı

- Trained a passive-dynamic biped via FSM → behavior cloning → PPO (JAX/Equinox/Optax; Brax/MuJoCo)
- Built expert controllers and a GPU-parallel PPO pipeline; achieved stable gait within $\sim 10^5$ steps
- Released reproducible code, configs, and logs; ran large hyperparameter sweeps for robust policies

SURover (Sabancı University Rover Team), Technical Team Captain Jun 2022–Jan 2025

- Built a ROS+Gazebo digital twin; authored URDF/Xacro and MoveIt pipelines for arm and drivetrain
- Implemented multi-sensor fusion and ArUco-based fiducial SLAM with ZED2, RealSense D435i, and RPLIDAR-A3
- Developed a Wi-Fi joystick and GUI with watchdog and E-stop; led mechanical fabrication/cabling and Arduino-based motor control for a field-ready rover

Cart-Pole Swing-Up Control (JAX & MuJoCo) Oct 2024–Jan 2025

Supervisor: Dr. Aykut Cihan Satıcı

- Classical (LQR) and neural controllers with differentiable simulation (JAX/Equinox/Diffrax) in MuJoCo; robust swing-up under disturbances

Robotic Item Placement on Cluttered Desks (Baxter) Jan 2023–Jan 2024

Supervisors: Prof. Volkan Patoğlu, Prof. Esra Erdem

- Gazebo + ROS Noetic pipeline; resolved MoveIt integration for Baxter and demonstrated clutter-aware placement policies

HONORS & AWARDS

- Merit-Based Full Tuition Scholarship — Sabancı University 2020–Present
- Dean's High Honor — Sabancı University 2020–Present
- Ranked 7th in the Electronics Engineering Department — Sabancı University Present
- Graduated 1st in the Mechatronics Engineering Department — Sabancı University 2025
- Extracurricular Student Activities Leadership Award — Sabancı University 2023, 2024

TECHNICAL SKILLS & LANGUAGES

Robotics Stack:	ROS 1/2 (Noetic, Humble); URDF/Xacro; ros2_control; MoveIt/MoveIt2; Navigation2; Gazebo
Modeling & Control:	State estimation (EKF); LQR; MATLAB/Simulink; Autolev
Learning & Simulation:	RL (BC, PPO; OpenAI Gym); JAX (Equinox, Optax, Diffrax); MuJoCo; Brax
Perception:	OpenCV; MediaPipe; ArUco; RealSense SDK
Programming & Dev:	Python; C++; Docker; Git
Embedded & Electronics:	Arduino; ESP32; LTSpice
Languages:	English (Advanced); Turkish (Native)

ACTIVITIES & VOLUNTEERING

Sabancı IEEE Student Chapter Istanbul, Turkey

Board Member Sep 2022–Sep 2024

- Oversaw SURover team operations, leadership coordination, and project continuity within the chapter

Sabancı University SIAM Student Chapter Istanbul, Turkey

Financial Affairs Coordinator Sep 2022–Jun 2024

- Managed budgets and sponsorships; organized seminars, reading groups, and on-campus math competitions

Civic Involvement Projects (CIP) Istanbul, Turkey

Volunteer Apr 2021–Jun 2021

- Narrated and edited audiobooks for visually impaired listeners to improve accessibility